1. Suppose you are playing an old video game where the player has to navigate a maze as given below. The player starts at the position labeled S and needs to reach the position labeled G. You can move up, down, left and right. However, your controller is old and unreliable as a result 10% of time it just moves in a random direction irrespective of the direction pressed by you. On reaching the goal state you get a reward of +1. Any action that leads to a collision with the grid walls will give -1 reward and the position of player does not change. All other rewards are zero.

a	b	G
c	d	e
S	f	g

You observe an episode which has the following succession of state {S,c,c,d,b,G}. The actions taken by the agent were {UP, RIGHT, RIGHT, UP, RIGHT}. The agent repeats the state c due to collision with the grid wall.
If we implement Q-Learning with **α**=0.9, then what will be the value of Q(c, RIGHT)? All Q(s,a) pairs are initialized to zero. Assume discounting factor to be 1. Round off the answer to two decimal points, e.g. if your answer is 0.1284 then write 0.13
Ans. -0.09

Q(c, RIGHT) will be updated twice.

At first update, since a collision happens a reward of -1 will be received.
Q(c, RIGHT) = (1-**α**).Q(c, RIGHT) + **α**(R+Q(c, RIGHT))
= 0.1x0 + 0.9(-1)
= -0.9

At second update
Q(c, RIGHT) = (1-**α**).Q(c, RIGHT) + **α**(R+Q(d, UP))
= 0.1x(-0.9) + 0.9(0)
= -0.09

2. Consider the same setting as the first question. You observe the same episode as above and you are doing Q-Learning. If we have epsilon greedy policy with epsilon=0.2 then what is the probability of agent taking action RIGHT in state C after the first episode is over?
   Round off the answer to two decimal points, e.g. if your answer is 0.1284 then write 0.13
   Ans. 0.05

   Q(c,RIGHT)= -0.09
   Q(c,UP)=Q(c,DOWN)=Q(c,LEFT)=0
   Epsilon greedy policy will pick any one of UP, DOWN, LEFT as the action.
   Q(c,RIGHT) action will only be taken using epsilon/4 probability.

3. If we follow Boltzmann exploration strategy, then how should the temperature (T) parameter be varied?
   a. Start off with low value and increase it gradually
   b. Start off with high value and decay it gradually
   c. Keep it fixed at a small value

   Initially we prefer to do exploration as we are yet to identify a good policy but over time we need to exploit more to gain from the good policy we have found during exploration. Higher temperature leads to more exploration whereas lower temperature encourages greedy action selection. Thus, we should start with high temperature and decay it gradually.

4. Which of the following actions can lead to a change in the policy learned by an RL agent which follows the epsilon-greedy exploration scheme?
   a. Change in transition probability function.
   b. Change in the discount factor
   c. A positive constant is multiplied to all the rewards

   Since Q(s,a) values depend on the transition probability function, so if transition probability function is changed Q values would change, hence the policy can change.

   If the discount factor is small then policies that optimize will be preferred over policies that maximize long term reward, whereas if the discount factor is close to one then policies optimizing long term reward will be preferred. Thus, a change in the discount factor may lead to change in policy learnt.

   If all rewards are scaled by a positive constant than all Q(s,a) values would change by the same constant, thus there relative ordering wil not change. As a result argmax_a Q(s,a) will not change for any state. Thus optimal policy learnt stays the same.

5. How many parameters are needed to be estimated during Q-Learning in a world with S states each having A different actions?
    a. **O(S.A)**
    b. O(S)
    c. O(A)
    d. $O(A^S)$

    For Q-Learning we need to estimate Q(s,a) values for all the states and actions. There are no other parameters than need to be learnt. So Q-Learning needs to estimate only O(S.A) parameters.

6. Agent's environment can be of two types: Episodic and Continuous.
    Episodic tasks are the tasks that have a terminal state.
    Continuous tasks are the tasks that don't have a terminal state.
    Which of the following statement(s) is/are correct for Temporal Difference Learning (TD) and Learning using Empirical Estimation (EE)?
    a. **TD can learn in a continuous environment.**
    b. **EE can learn in an episodic environment.**
    c. TD is guaranteed to converge to the optimal value given infinite data independent of the sequence of learning rates.
    d. **EE is guaranteed to converge to the optimal value given infinite data.**

    TD can learn in a continuous environment because it only requires 1 step by the agent to do an update. It can do that infinitely to learn a good policy hence option A is correct. EE can learn in an episodic environment as we have already done in the lectures hence option B is correct. The TD learning algorithm is highly dependent on the sequence of learning rate and initial values. The learning rate is required to decrease appropriately for TD to converge hence option C is incorrect. EE converges to the optimal value for infinite training data as it does not have any parameters to tune and it uses good estimates to learn the value function for a policy using the rewards seen during interaction with the environment hence option D is correct.

7. Consider the following experience sampled in an environment for a policy p.
    Format: (State, Reward)
    Sample 1: (A, 0), (B, 1)
    Sample 2: (B, 1)
    Sample 3: (B, 1)
    Sample 4: (B, 0)
    Sample 5: (B, 1)
    Sample 6: (B, 1)
    Sample 7: (B, 1)
    Sample 8: (A, 0), (B, 0)

Using Empirical Estimates, find the value $V^p(A)$. (up to 2 decimal places)

Ans: 0.5

State A is seen in two samples 1 and 8.
In sample 1, reward earned after encountering state A = 0 + 1 = 1.
In sample 8, reward earned after encountering state A = 0 + 0 = 0.
$V^p(A) = (1 + 0) / 2 = 0.5$

8. Consider the following experience sampled in an environment for a policy p.
   Format: (State, Reward)
   Sample 1: (A, 0), (B, 1)
   Sample 2: (B, 1)
   Sample 3: (B, 1)
   Sample 4: (B, 0)
   Sample 5: (B, 1)
   Sample 6: (B, 1)
   Sample 7: (B, 1)
   Sample 8: (A, 0), (B, 0)

   If the above samples are continuously fed to a Temporal difference learning algorithm (i.e. the algorithm trains on the same set of above samples infinitely), what is the value $V^p(A)$ found by the algorithm? (up to 2 decimal places). Assume no discounting.

   Ans: 0.75

   Assuming that the TD algorithm is run with appropriate learning rates for convergence.

   Intuitively, the TD algorithm will believe the following:
   Under policy p, the action in state A always leads to the state B with a reward of 0.
   Under policy p, the action in state B always leads to the terminal state, 75% of the times yield a reward of 1 and 25% of the times yielding a reward of 0.

   E() denotes Expected Value and T denotes terminal state.
   Therefore, $V^p(A) = E(R_{AB} + V^p(B)) = E(R_{AB} + R_{BT} + V^P(T)) = E(R_{AB} + R_{BT})$
   $= E(R_{AB}) + E(R_{BT}) = (1 * 0) + (0.75 * 1 + 0.25 * 0) = 0.75.$

9. Which of the following statement(s) is/are correct for Model-based and Model-free reinforcement learning methods?
   a. **Model-based learning requires more parameters and data to learn.**
   b. Model-free learning can exploit the underlying MDP structure.

c. Model-free learning can simulate new episodes from past experience.

Model Based requires parameters for learning the transition function of the MDP hence option A is correct. Model free learning does not work with any transition function and just collects samples to learn hence it cannot exploit the underlying MDP structure unlike Model based learning hence option B is incorrect. Model free learning does not maintain the transition probabilities so cannot simulate new episodes from past experience unlike Model based learning hence option C is incorrect.

10. Which of the following strategies are favorable methods for managing exploration in the Q-learning algorithm?
    a. **Random actions with probability p and greedy action with probability 1-p.**
    b. Performing actions that have been performed less with a lesser probability.
    c. **Maximizing exploration functions where an exploration term is added to the exploitation term**

Choosing random actions with probability p and greedy action with probability (1-p) is exactly the same as the epsilon-greedy policy, a classic method for exploration hence option A is correct. Performing actions that have been performed less with lesser probability is contrary to exploration as those actions won't be perfectly understood by the agent hence option B is incorrect. Maximizing exploration functions where an exploration term is added to the exploitation term would help in managing exploration because it would try to balance both exploration and exploitation in order to maximize hence option C is correct.